UNIVERSITY OF
South Carolina

Molinaroli College of
Engineering and Computing

Spring 2026

# EMCH792 – Optimal State Estimation

## Homework 1

Submission deadline: Wednesday, 01/28/2026, 11:59pm EST

- Submission only through Blackboard, email submissions will not be accepted.
- Make sure you properly upload the file(s) and that you can successfully view your submission on BB. Keep a copy of the submission receipt you get from BB.
- Late submissions will be graded with the policy described in the syllabus.
- For questions email tsucin@email.sc.edu and copy vitzilaios@sc.edu

# Homework Instructions

## Written Portion

Solve the following problems (refer to the textbook or Appendix of this assignment) by hand and provide your solutions in a scanned PDF document.

I. Exercise 1.2, pages 45-46
II. Exercise 1.3, page 46
III. Exercise 1.4, page 46
IV. Exercise 1.6, page 46
V. Exercise 1.7, page 46

## Computer Portion

This is an introductory homework to practice **MATLAB** and linear algebra using **MATLAB**. All the files that you will need for this homework are in the **ws** folder. You are expected to work inside this folder only. Open **MATLAB** and set the current folder to **ws**, and do the following:

1. Start by opening the live script file <u>"HW_1_manual.mlx"</u> using **MATLAB**. This will guide you through the basic concepts like initializing vectors and matrices as well as some simple operations like addition and subtraction. Read through the file and understand how each function is implemented in **MATLAB**.
2. Open the file <u>"HW_1.m"</u>. This is a normal script and not a live script. If you open it as a live script, the instructions might appear messy. <u>Moreover, we won't be able to provide any direct feedback on BB if you submit it as a live script.</u> Start by editing the variable **lastName**. You need to answer the questions in the comments by editing the file and adding your code below each comment.
   a. When the question asks you to show that two arguments a and b are equal use the provided function <u>"checkEqual(a, b)"</u>.
      i. If the arguments are equal, the function output is <u>"The two arguments are equal"</u>.
      ii. If they are not equal, the function outputs <u>"The two arguments are NOT equal"</u>.
   b. When a question asks you to find\show the rank of a matrix or the output of an operation you can either just show the answer immediately (e.g. rank(A)) or push the answer in a variable and show it (e.g. rank_A = rank(A)).

## Homework Submission

Your submission must include the following files:

- **PDF document with the scanned solutions of the written portion**
- **"HW1.m" -> The edited MATLAB script with your solutions**

**Upload both files on Blackboard. You are responsible to verify that you submitted the correct files.**

Make sure that the scripts you submit can run. Don't copy the results that appear in the command window in your scripts and make sure that if you add any comments in your scripts that they follow the proper **MATLAB** syntax. **Scripts that do not run will not be fully graded.**

## Rubric

| **Problem** | **Points** |
|---|---|
| **Written portion** | |
| **I** | **10** |
| **II** | **10** |
| **III** | **10** |
| **IV** | **10** |
| **V** | **10** |
| **Computer portion** | |
| **Problem 1** | **10** |
| **Problem 2** | **15** |
| **Problem 3** | **25** |

## Appendix

**1.2** Find two $2 \times 2$ matrices $A$ and $B$ such that $A \neq B$, neither $A$ nor $B$ are diagonal, $A \neq cB$ for any scalar $c$, and $AB = BA$. Find the eigenvectors of $A$ and $B$. Note that they share an eigenvector. Interestingly, every pair of commuting matrices shares at least one eigenvector [Hor85, p. 51].

**1.3** Prove the three identities of Equation (1.26).

**1.4** Find the partial derivative of the trace of $AB$ with respect to $A$.

**1.6** Suppose that $A$ is invertible and

$$\begin{bmatrix} A & A \\ B & A \end{bmatrix} \begin{bmatrix} A \\ C \end{bmatrix} = \begin{bmatrix} 0 \\ I \end{bmatrix}$$

Find $B$ and $C$ in terms of $A$ [Lie67].

**1.7** Show that $AB$ may not be symmetric even though both $A$ and $B$ are symmetric.